IVDA - Homework report

Catrina Bodean 25-752-783

May 2026

1 Introduction

2 Background and related works

Physical activity has been shown to influence the progression of chronic diseases: increased activity tends to delay the onset of disease or slow down its progression, while sedentarism tends to increase the likelihood of developing a chronic condition. (Fang et al., 2026; Poleszak et al., 2026) Not only that, but the patients' mood and comorbidities like depression and anxiety are also affected, being easier to manage for people who are more active than for those who aren't. (Stephens et al., 2019). However, physical activity is not something that exists in a vacuum: aside from the disease itself, it can also be influenced by external factors like temperature, humidity, or air pressure. (Ramadan et al., 2024; Zhang et al., 2025).

As wearable sensor devices have become near ubiquitous and fairly accessible, researchers now have the opportunity to track patients in a more natural environment, for longer periods of time (Rekeland et al., 2022; Zhang et al., 2025). Thanks to the vast amounts of data generated, this presents the unique chance to better explore the influence of external factors on physical activity and to quantify it with "hard numbers" rather than subjective self-reports.

3 Methods

All hypotheses have been tested for a subset of 43 participants: participant with ID 4063 has been excluded, because their recorded wearable data spanned less than 48 hours. Similarly, the intervention and follow-up phases have been ignored due to missing information on when they took place.

3.1 Hypothesis 1: Physical activity is related to weather

Based on Zhang et al., 2025, the first hypothesis I tested was that the physical activity of the patients was influenced by the weather; more specifically, warm (but not hot) weather with few precipitations would correlate to a higher number of steps than cold or hot weather and higher quantity of precipitation. To test it, I used historical data from MeteoSwiss (see 4 for details), specifically daily maximum temperature and precipitation. I calculated the total number of steps a patient took for every measured day, retrieved the weather information for the corresponding days, and plotted the number of steps and the maximum temperature, respectively amount of daily precipitation. I conducted a visual analysis of the information and corroborated it using pandas' correlation function for each parameter.

3.2 Hypothesis 2: The general level of activity is influenced by age and disease stage

The second hypothesis I tested was if the general activity level of the patients' was directly related to their age and the progression stage of their disease, namely if younger people or those who are in the early stages are more physically active than older people an/or people who are in the later stages. I measured the average number of steps per day for each patient and calculated the correlation between this number and age, respectively disease type, and a multivariate correlation between steps and both age and disease type.

4 Data characterization

I used three datasets:

1. clinical markers: 44 records containing the patients' ID, year of birth, sex, and disease type (early stage, late stage, fast progressing)
2. wearable data: a set of 44 files containing the hourly measured number of steps, minutes of sleep and average hourly heartbeat for each separate patient
3. historical meteorological data: I downloaded a dataset from MeteoSwiss[1] containing all the recorded meteorological data at the Zurich Fluntern meteorological station; from this I extracted a subset spanning only the year 2021 and containing the daily maximum air temperature measured at 2 metres above ground in Celsius and the average daily precipitation from 0 to 23:59 measured in millimetres (presumably per square metre).

5 Implementation and sofware use

The entire practical work was conducted using the Python programming language and the Google Colab programming environment. I used the following Python packages:

- reading files: google.colab and os
- data analysis: pandas, numpy, statsmodels.api
- plotting: matplotlib.pyplot, plotly.graph_objects, plotly.subplots, seaborn, plotly.express

6 Results

6.1 Hypothesis 1

Generally speaking, temperature and precipitation have little influence on the physical activity of a patient. During the visual analysis, I didn't observe any obvious trends, and the correlation coefficients support this finding. While there are a few patients for whom temperature is a significant factor for movement, overall the correlation factor is quite weak (tables 1 and 1).

Figures 6.1 and 6.2 show a comparison between the number of steps and the maximum temperature or the total precipitation for 5 patients. The graphs for all 43 patients are represented in figures 6.4 and 6.3.

[1] MeteoSwiss Open Data Explorer

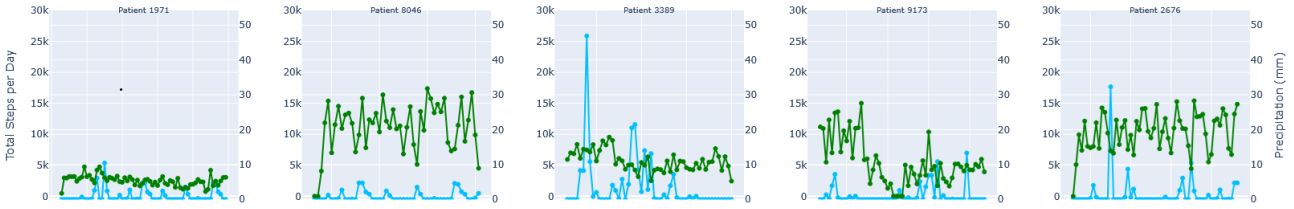


Figure 6.1: Total number of steps (green) and total precipitation (blue) per day for 5 patients



Figure 6.2: Total number of steps (green) and maximum temperature (maroon) per day for 5 patients

Table 1: Correlation factor between daily steps and maximum temperature

Patient_ID	Correlation Steps–Temperature
7996	-0.6013
2909	-0.5105
6661	-0.4016
9173	-0.2870
4457	-0.2060
7458	-0.2048
8046	-0.1907
2676	-0.1879
3389	-0.1383
7183	-0.1377
9411	-0.1301
7513	-0.0781
9085	-0.0331
6947	-0.0055
6155	0.0094
8826	0.0156
7928	0.0244
2854	0.0280
4420	0.0653
2130	0.0755
2815	0.1127
6395	0.1233
1120	0.1365
5441	0.2074
2589	0.2136
6823	0.2372
5338	0.2393
5977	0.2403
4441	0.2474
5479	0.2637
6864	0.2646
7016	0.2657
9501	0.2764
3405	0.2912
1971	0.3619
1556	0.3685
4475	0.3704
8446	0.3756
4701	0.3985
4349	0.4910
8388	0.6066
8147	0.6580

Table 2: Correlation factors of daily steps and precipitation

Patient_ID	Correlation Steps-Precipitation
5441	-0.4045
7928	-0.3506
2589	-0.3327
9501	-0.2968
4475	-0.2908
1556	-0.2856
9411	-0.2799
7016	-0.2369
2854	-0.2136
3425	-0.1799
9085	-0.1842
2676	-0.1720
9926	-0.1728
5479	-0.1462
8446	-0.1440
6155	-0.0980
4349	-0.0960
8147	-0.0958
4701	-0.0807
6823	-0.0778
8046	-0.0531
7513	-0.0469
6947	-0.0411
6395	-0.0274
1120	-0.0194
4457	-0.0127
7183	-0.0090
6864	-0.0049
4420	0.0052
3389	0.0097
9173	0.0259
4441	0.0459
2130	0.0486
1971	0.0493
7458	0.0521
5977	0.0586
5338	0.0642
8826	0.0711
8388	0.1355
2815	0.1562
2909	0.1840
6661	0.2121
7996	0.2317



Figure 6.3: Total number of steps (green) and total precipitation (blue) per day for all 43 patients



Figure 6.4: Total number of steps (green) and maximum temperature (maroon) per day for all 43 patients

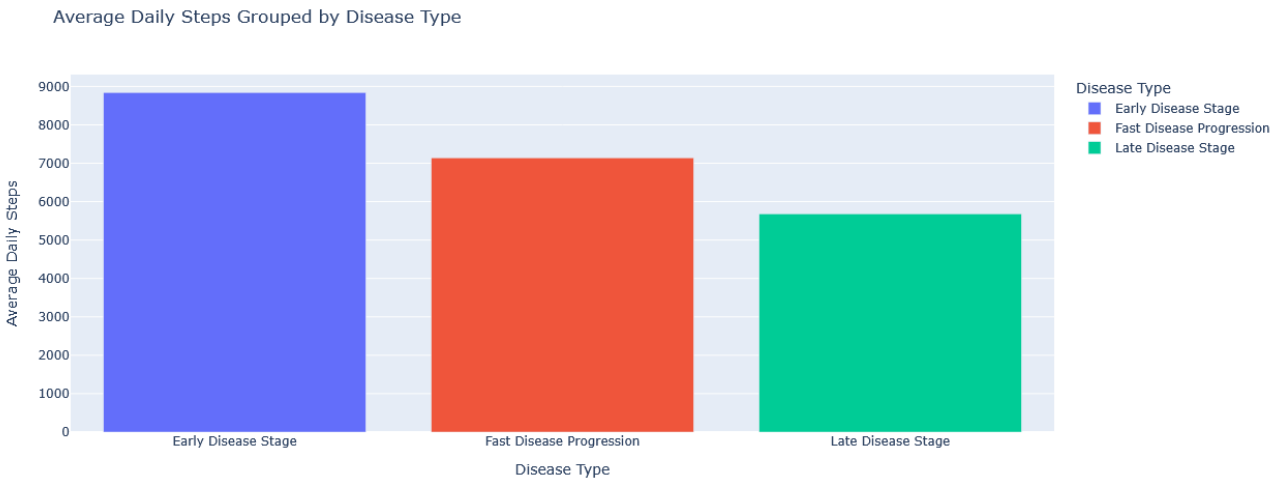


Figure 6.5: Average number of average daily steps per disease type

6.2 Hypothesis 2

At first, grouping the average number of steps by disease type and then calculating the average for each group (figure 6.5) suggests that people in the early disease stage move more than those in the late stage, which is to be expected. However, creating a scatterplot of each patient's average daily number of steps, age, and disease type reveals that there is no clear clustering of the data (figure 6.6). The individual correlation factor shows a weak correlation between steps and age, respectively disease type (figures 6.7 and 6.8). When calculating the multivariate correlation, I obtained an **R-squared value of 0.2205** and a **multiple correlation coefficient of 0.4695**; this suggests that age and disease type have a weak to moderate influence on a patients amount of physical activity.

7 Discussion

The results are surprising, as, contrary to the literature, there is no clear significant correlation between physical activity and weather or disease progression. However, the approach was not thorough, lacking statistical analyses or deeper exploration.

8 Limitations

The most important limitation is, in my opinion, the number of patients that had their data recorded. I believe a cohort of 44 people is not big enough to lead to statistically significant results, but rather it can be used as a starting point for bigger and more thorough experiments.

Another limitation is the limited data on the patients' lifestyle. It is mentioned in the homework description that the patients have spent a period of time in a specialised rehabilitation facility, after which they went home and continued to wear the Fitbit bracelets. However, there is no indication in the data provided when this move took place; as people tend to be more serious about physical therapy in a controlled environment, but then be less rigorous at home or other non-controlled environments, it is possible the results are not completely relevant. For

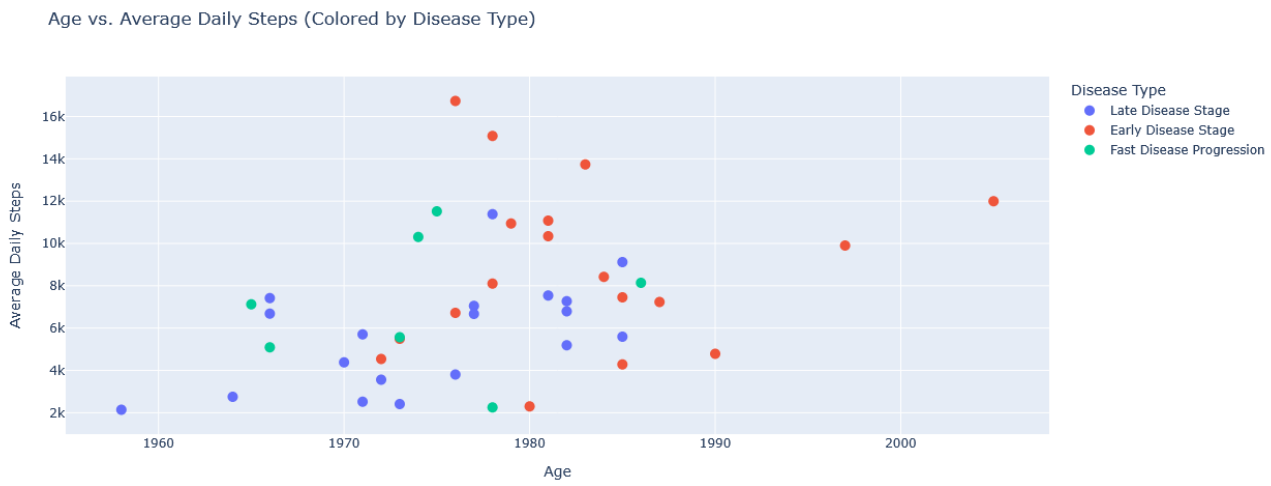


Figure 6.6: Distribution of disease type and number of steps by age

similar reasons it might also help knowing whether the patients have support at home or not.

Finally, the patient measurements were taken in 2021, during the SARS-CoV-2 pandemic. As such, it is possible that the measured data was influenced by whether or not the patient had had SARS-CoV-2 or whether they were administered the vaccine, or that the patients' behaviour was influenced by the pandemic and the public health measures in place (e.g., they quarantined themselves in order to avoid getting infected). However, this information is not available, but it might significantly alter the results.

9 Declaration of AI use

I used Google Colab's generative AI engine Gemini to generate the code and the plots I used in this homework, as well as suggested auto-completions for the code I wrote myself. Since I have previous experience with pandas, numpy, matplotlib and seaborn, I decided it would be faster for me to review and correct the generated code than write it by hand. The hypotheses, methods of evaluation, and results were decided and interpreted respectively by me with no input from Gemini.

I also used Overleaf's auto-complete, spell check, code check, and auto-bracket-close functions when writing this document.

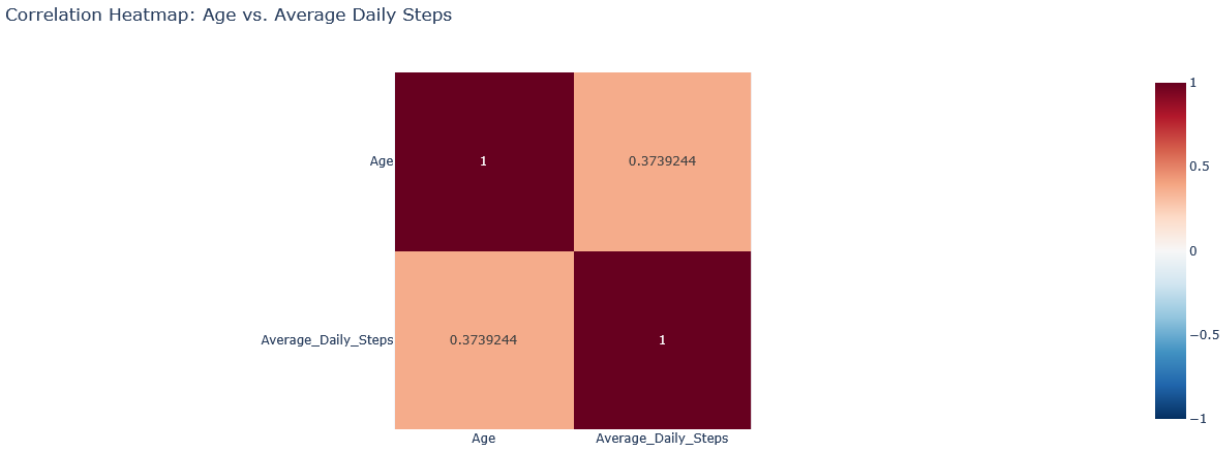


Figure 6.7: Heatmap of correlation between age and average number of daily steps



Figure 6.8: Heatmap of correlation between age and average number of daily steps



Figure 6.9: Representation of daily step total (green) and daily maximum temperature (maroon)

References

Fang, Z., Wang, P., Rosner, B., Giovannucci, E., & Song, M. (2026). Sustained physical activity offers benefits beyond activity volume in chronic disease prevention. *Nature Communications*, *17*. https://doi.org/10.1038/s41467-026-69552-4

Poleszak, J., Łuba, Z., Tyszko, K., Rusek, K., Pikur, K., Górska, A., Błażukiewicz, J., Balicki, J., NIDA, A., & Posturzyńska, A. (2026). Lifestyle, diet, and physical activity in the prevention and rehabilitation of chronic diseases: A review. *Wiadomości Lekarskie*, *79*, 518–524. https://doi.org/10.36740/WLek/217892

Ramadan, G., Samin, G., Haryanto, A., Kadir, S., & Okilanda, A. (2024). A cross-sectional study: The impact of extreme weather, sleep quality, and physical activity. *Retos*, *60*, 968–973. https://doi.org/10.47197/retos.v60.109307

Rekeland, I., Sørland, K., Bruland, O., Risa, K., Alme, K., Dahl, O., Tronstad, K. J., Mella, O., & Fluge, Ø. (2022). Activity monitoring and patient-reported outcome measures in myalgic encephalomyelitis/chronic fatigue syndrome patients. *PLOS ONE*, *17*, e0274472. https://doi.org/10.1371/journal.pone.0274472

Stephens, S., Shams, S., Lee, J., Grover, S., Longoni, G., Berenbaum, T., Finlayson, M., Motl, R., & Yeh, E. (2019). Benefits of physical activity for depression and fatigue in multiple sclerosis: A longitudinal analysis. *The Journal of Pediatrics*, *209*. https://doi.org/10.1016/j.jpeds.2019.01.040

Zhang, Y., Folarin, A., Ranjan, Y., Cummins, N., Rashid, Z., Conde, P., Stewart, C., Sun, S., Vairavan, S., Matcham, F., Oetzmann, C., Siddi, S., Lamers, F., Simblett, S., Wykes, T., Mohr, D., Haro, J., Penninx, B., Narayan, V., & Pratap, A. (2025). Assessing seasonal and weather effects on depression and physical activity using mobile health data. *npj Mental Health Research*, *4*. https://doi.org/10.1038/s44184-025-00125-x